Food and Chemical Toxicology 179 (2023) 113984

Contents lists available at ScienceDirect

Food and Chemical Toxicology

journal homepage: www.elsevier.com/locate/foodchemtox

# Impact of food matrices on the characteristics and cellular toxicities of ingested nanoplastics in a simulated digestive tract

Zhiming Li[a], Yuji Huang[a], Yizhou Zhong[a], Boxuan Liang[a], Xingfen Yang[b], Qing Wang[c], Haixia Sui[d,**], Zhenlie Huang[a,*]

[a] NMPA Key Laboratory for Safety Evaluation of Cosmetics, Guangdong Provincial Key Laboratory of Tropical Disease Research, Department of Toxicology, School of Public Health, Southern Medical University, Guangzhou, 510515, China
[b] Food Safety and Health Research Center, School of Public Health, Southern Medical University, Guangzhou, 510515, China
[c] Department of Toxicology, School of Public Health, Guangdong Provincial Key Laboratory of Food, Nutrition and Health, Sun Yat-sen University, Guangzhou, 510080, China
[d] NHC Key Laboratory of Food Safety Risk Assessment, China National Center for Food Safety Risk Assessment, Beijing, 100022, China

ARTICLE INFO

Handling Editor: Dr. Bryan Delaney

*Keywords:*
Microplastics and nanoplastics
Food matrix
Gastrointestinal simulated model
Combined effect
Cadmium
Risk assessment

ABSTRACT

Microplastic and nanoplastic (MNP) pollution has become a major global food safety concern. MNPs can interact with food matrices, and their passage through the gastrointestinal tract can modify their properties. To explore whether and how food matrices influence MNP toxicity, we investigated the interactions between polystyrene nanoplastics (PS-NPs) and food matrices, using an *in vitro* gastrointestinal digestion model. Then, we tested cell viability, particle uptake and cellular toxicities induced by PS-NPs with food matrices in Caco-2 cells. The results showed that PS-NPs were aggregated, both with and without food matrices, after *in vitro* gastrointestinal digestion. Glyceryl trioleate exerted greater ability to stabilize digestas and to disperse PS-NPs than starch and bovine serum albumin. The protein corona's protein composition on PS-NPs varied when it interacted with different food matrices. Moreover, when combined with food matrices, the PS-NPs' uptake was enhanced, thus aggravating cellular inflammation, stress, and apoptosis levels. Finally, through co-exposure to a mixture of food matrices, we found a combined negative effect of PS-NPs and cadmium on cellular inflammation, stress, and apoptosis levels. This is the first study to compare the impact of various food matrices on the characteristics and cellular toxicities of ingested NPs in a simulated digestive tract.

## 1. Introduction

Microplastic and nanoplastic (MNP) pollution has drawn the attention of researchers from all walks of life. MNPs' presence in human biosamples, including stool (Zhang et al., 2021), blood (Leslie et al., 2022), lung tissue (Amato-Lourenco et al., 2021), placentas (Ragusa et al., 2021) and thrombi (Wu et al., 2022), has indicated these particles' potential threat to human health. MNPs have been found in edible animal species (seafood and chicken) and food samples such as salt, sugar, honey, milk, fruit, soft drinks and drinking water (Pironti et al., 2021), suggesting that the digestive tract is a primary pathway of human exposure to MNPs. It has been estimated that humans ingest roughly 0.1–5 g of MNPs per week through food and beverage consumption (Cox et al., 2019). Additionally, MNP concentration in the feces of inflammatory bowel disease patients (41.8 items/g) is higher than that of healthy individuals (28.0 items/g) (Yan et al., 2021). An experimental study has also demonstrated MNP biodistribution and accumulation in mouse intestines, and thereafter that it induces reactive oxygen species-mediated epithelial cell apoptosis and impairs the gut barrier (Liang et al., 2021). Considering the escalating exposure and accumulating evidence of damage associated with MNPs, there is an urgent need for comprehensive investigations into MNP-induced intestinal toxicity. Moreover, nanoplastics (NPs) arouse greater intestinal toxicity than microplastics (MPs) due to their greater capacity to penetrate the body (Li et al., 2021; Liang et al., 2021).

The European Food Safety Authority has produced guidance with recommendations for risk assessment for nanomaterials in the food

* Corresponding author. Department of Toxicology, School of Public Health, Southern Medical University, 1023-1063 Shatai Nan Road, Guangzhou, 510515, China.
** Corresponding author.
*E-mail addresses:* suihaixia@cfsa.net.cn (H. Sui), huangzhenlie858252@smu.edu.cn (Z. Huang).

https://doi.org/10.1016/j.fct.2023.113984
Received 17 June 2023; Received in revised form 27 July 2023; Accepted 7 August 2023
Available online 9 August 2023

**Abbreviations**

| | |
|---|---|
| BSA | bovine serum albumin |
| CASP | caspase |
| CCK-8 | cell counting kit-8 |
| Cd | cadmium |
| DLS | dynamic light scattering |
| DMEM | Dulbecco's modified Eagle medium |
| FBS | fetal bovine serum |
| FFM | fasting food model |
| GF | green fluorescence |
| GSTP1 | glutathione S-transferase P1 |
| GT | glyceryl trioleate |
| HCl | hydrochloric acid |
| HO1 | heme oxygenase-1 |
| HSP70 | heat shock protein 70 |
| IL | interleukin |
| MNP | microplastic and nanoplastic |
| MPs | microplastics |
| NF | non-fluorescence |
| NPs | nanopalstics |
| PBS | phosphate buffered saline |
| PDI | polymer dispersion index |
| PS | polystyrene |
| qPCR | quantitative polymerase chain reaction |
| SD | standard deviation |
| SDS-PAGE | sodium dodecyl sulfate polyacrylamide gel electrophoresis |
| SEM | scanning electron microscope |
| SFM | standard food model |
| SOD | superoxidedismutase |

chain (Hardy et al., 2018). The guidance outlines a structured approach for testing nanomaterials for identification and characterization of toxicological hazards (Hardy et al., 2018). It also highlights the importance of following the fate of nanomaterials in the gastrointestinal tract to determine whether the particles consist of primary particles only, or could also consist of aggregates and agglomerates during the digestive process (Hardy et al., 2018). *In vitro* models of the gastrointestinal tract might be useful for investigating NP alterations in food after ingestion. In addition to providing a solution to the issue of ethical restrictions, and being more cost-effective and expedient than *in vivo* studies, *in vitro* models facilitate the collection of large samples, the realization of various controls, and obtaining reproducible results (Hardy et al., 2018). Even though the approach is reductionist rather than holistic, these models can explicate the mechanisms at play, and might be useful for regulatory authorities in assessing various nanomaterials' safety. Thus, *in vitro* gastrointestinal models are crucial to investigate the effects of human gastrointestinal tract on the MNPs, allowing for the simulation of real-world scenarios.

The Caco-2 cell model has been widely used to observe the intestinal toxicity of MNPs once they have been ingested (Domenech et al., 2021; Gautam et al., 2022; Stock et al., 2021; Wu et al., 2020). For instance, MNPs have been found to inhibit the proliferation of Caco-2 cells in both time- and dose-dependent manners (Tolardo et al., 2022; Wu et al., 2020). Concomitantly, it has been documented that oxidative stress and inflammation are associated with MNP-induced toxicity (Gautam et al., 2022; Tolardo et al., 2022; Wu et al., 2019, 2020). These studies have been performed with pristine-MNPs to study MNPs' impact on intestinal cells. *In vitro* models of the gastrointestinal tract have also been used to investigate MNPs' fate after they have been ingested (DeLoid et al., 2021, 2022; Liu et al., 2020a). Nevertheless, these studies have under-emphasized the presence of food matrices. A growing body of literature has reported that various food matrices can alter nanoparticles' behavior (Laloux et al., 2020; Lichtenstein et al., 2015; Zhou et al., 2021). Since nanoparticles' toxic effects are related to their physicochemical properties, any alterations that transpire before nanoparticles reach the intestinal cells may affect their potential toxicity. Therefore, the interaction between food matrices and MNPs should be included in cytotoxicity investigations to realize accurate risk assessment.

To explore the impact of food matrices on the NP gastrointestinal fate and NP-induced intestinal enterocyte toxicity, we performed *in vitro* gastrointestinal digestion on 50 nm polystyrene (PS)-NPs with three kinds of food matrices—starch, bovine serum albumin (BSA) and glyceryl trioleate (GT). Digested PS-NPs' characteristics were disclosed by dynamic light scattering (DLS), scanning electron microscope (SEM) and protein corona analysis. Then, we evaluated the effects of various food matrices on PS-NP-induced enterocyte toxicities, including outcomes such as cellular viability, particle uptake, and mRNA levels in genes related to inflammation, stress and apoptosis, in Caco-2 cells. Finally, we discovered the combined effects of PS-NPs and cadmium (Cd) with a mixture of food matrices. This study will provide a key piece in the puzzle of NP-induced intestinal toxicity, and will facilitate risk assessment of NPs in the food chain.

## 2. Materials and methods

### 2.1. Chemicals and reagents

We purchased the 50 nm non-fluorescence (NF) and green fluorescence (GF) PS-NP suspensions from Magsphere (Pasadena, CA, USA). These PS-NPs' characterization has been described in our previous studies (Liang et al., 2021, 2022a). GF PS-NPs were only used in cellular uptake assay, and NF PS-NPs were used for other investigations. We purchased other chemicals from Mclean Biochemical Technology Co., Ltd. (Shanghai, China), and purchased the biochemical reagents from Sigma-Aldrich (St. Louis, MO, USA).

### 2.2. In vitro gastrointestinal digestion

An overview of the study design is shown in Fig. 1. We used an *in vitro* gastrointestinal tract to simulate human NP intake. The model consisted of a three-step procedure, involving the oral cavity (2 min), stomach (2 h), and small intestine (2 h), which is available as a standard operating procedure (Bove et al., 2017). The simulated digestive juices were prepared in sterile conditions by combining salt solutions, organic compounds, enzymes and proteins to obtain the final concentrations in a total reaction volume of 50 mL, which has been reported in our previous study (Xu et al., 2021). The digestive process steps are as follows: the digestive juices are added in the order of human digestion. 120 μL salivary juice is mixed with 90 μL food input to imitate the oral cavity. The pH value is then adjusted to 6.5 ± 0.2 using 1 M hydrochloric acid (HCl), followed by shaking at 100 rpm and 37 °C for 2 min to simulate mastication. Next, the mixture is transferred to the simulated stomach. 260 μL gastric juice is mixed with the mixture, the pH value is adjusted to 1.5 ± 0.1 using 1 M HCl, and incubated for 120 min at 100 rpm and 37 °C. The mixture is then transferred into the simulated small intestine. Then, 240 μL duodenal fluid and 120 μL bile salts are added to the mixture, and the pH value is set at 7.8 ± 0.2 with 1 M sodium hydroxide. The mixture is then incubated for 120 min at 100 rpm and 37 °C and used as the final digestas. The entire digestive process is conducted under sterile conditions.

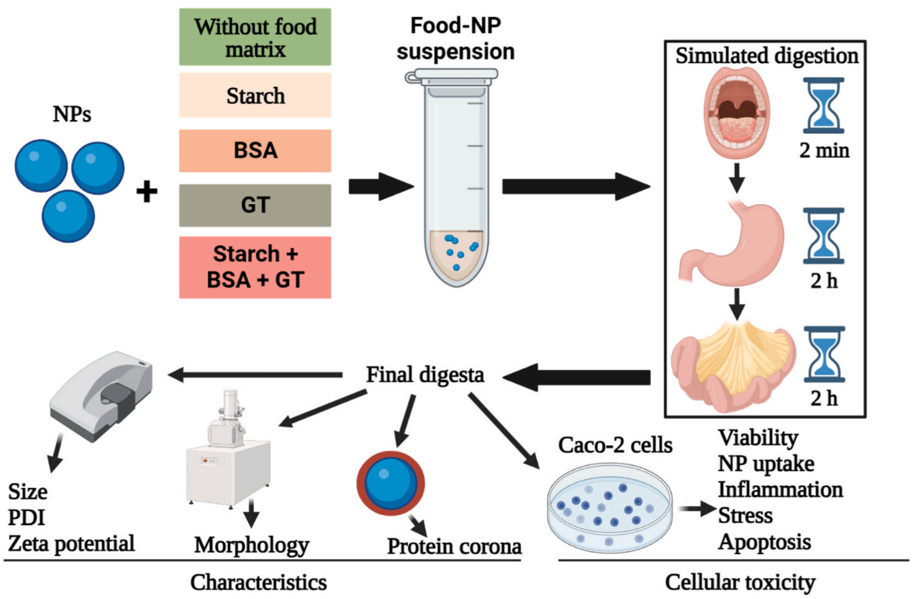


**Fig. 1.** Study design overview. Tested NPs with or without food matrices were subjected to simulated gastrointestinal digestion to mimic the physicochemical transformations that would occur *in vivo* prior to the interactions between ingested NPs and intestinal enterocytes. The final digestas containing the transformed NPs were characterized by DLS, SEM, and protein corona analysis. They were also applied to Caco-2 cells to assess NP uptake and to measure cellular toxic endpoints (cell viability, inflammation, stress, and apoptosis). BSA, bovine serum albumin; GT, glyceryl trioleate; NPs, nanoplastics; PDI, polymer dispersion index.

### *2.3. Food input*

The major constituents in most foods are water, proteins, carbohydrates, lipids, and minerals, but they may also contain various other minor constituents, such as surfactants, colors, flavors, buffers, acids, bases, preservatives, and nutraceuticals (Belitz et al., 2009). Among these constituents, carbohydrates, proteins and lipids comprise the vast majority of nutrients in food, and it has been suggested that they have the greatest impact on nanoparticle behaviors (Coreas et al., 2020; Qiaorun et al., 2022). Thus, we selected carbohydrates, proteins, lipids and a combination of these three macronutrients as food input in the simulated gastrointestinal digestion. Like previous studies (Hollebeeck et al., 2013; Laloux et al., 2020), starch, BSA and GT were chosen to represent carbohydrates, proteins and lipids, separately. According to European Union regulation No. 1169/2011 on the provision of food information to consumers, macronutrients' reference intakes are ≈260 g of carbohydrates, ≈50 g of proteins, and ≈70 g of lipids (Laloux et al., 2020). In the present study, by following a previous study (Laloux et al., 2020), we applied concentrations of 8.7 g/L starch, 5.6 g/L BSA and 7.8 g/L GT. These concentrations of three food matrices were used to represent the macronutrient concentrations found at the intestinal level by accounting for the volumes of ingested (2 L) and secreted fluids (7 L) along with the relative proportions of the major dietary nutrients.

Technical challenges in NP particle detection have led to insufficient human MNP exposure data. It has been estimated that most humans intake around 5 g of MNPs per week, primarily through drinking water (Senathirajah et al., 2021). Assuming an average fluid intake of 2 L/d, this would correspond to an average MNP intake of >350 μg/mL (DeLoid et al., 2021). Thus, in the present study, we chose 400 and 800 μg/mL as starting NP concentrations in food input. These NP concentrations were similar to the MNP doses applied in other studies with *in vitro* digestive tract models (DeLoid et al., 2021; Tan et al., 2020a). To ensure adequate nutrient delivery to cells for proper function and viability (DeLoid et al., 2017, 2021), the final small intestinal digesta was combined with complete 2 × Dulbecco's modified Eagle medium (DMEM) (Cienry, Hangzhou, China) containing 10% (v/v) fetal bovine serum (FBS, Gibco, CA, USA) and sterile water at a ratio of 1:2:1. This resulted in actual cell exposure concentrations with 10.84 and 21.68 μg/mL NPs, respectively. The actual cell exposure concentrations were used to describe the NP doses in the following investigations.

### *2.4. PS-NP characterization*

Without the DMEM, size, polymer dispersion index (PDI) and zeta potential of PS-NP suspensions in distilled water, both before and after *in vitro* gastrointestinal digestion, were obtained by a Zetasizer Nano ZS (Malvern Panalytical GmbH, Kassel, Germany). The morphology of PS-NPs was visualized using SEM (Zeiss sigma300, Carl Zeiss AG, Germany). With 10% 2 × DMEM, the pristine PS-NPs and the 4-fold diluted final digestas were used to measure the size, PDI and zeta potential to unveil the PS-NP characteristics in the cell exposure solution.

### *2.5. Analysis of protein corona on PS-NPs*

The treated PS-NPs were centrifuged for 30 min at 20,000 g (4 °C), and the supernatant was removed. The remaining pellet was resuspended with 1 mL phosphate buffered solution (PBS), and centrifugal operation was repeated. The wash procedure was repeated three times before resuspension of the final pellet with 60 μL radioimmunoprecipitation assay buffer. The supernatant was collected as hard corona previously absorbed on the surface of PS-NPs, and was quantified with a Bradford Protein Assay kit (Bio-Rad Laboratories, CA, USA). Then, the supernatant was denatured in 1 × SDS loading buffer at 100 °C for 10 min. Proteins were separated by 12% sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and visualized through coomassie brilliant blue R250.

### *2.6. Cell culture and treatment*

We cultivated Caco-2 cells (ATCC® HTB-37™, Manassas, VA, USA) in complete DMEM (Gibco) supplemented with 10% (v/v) FBS and 5% penicillin streptomycin double antibody (Gibco) in a humidified atmosphere of 5% $CO_2$ at 37 °C. The final small intestinal digesta was combined with complete 2 × DMEM containing 10% FBS and sterile water at a ratio of 1:2:1, followed by application to the cells. In order to make used complete DMEM comparable between groups, we diluted complete 2 × DMEM 2-fold with sterile water. The diluted complete 2 × DMEM was used in the pristine group. The cells were divided into the following 6 treatment groups. (1) Pristine group: cells were incubated in complete 2 × DMEM containing pristine PS-NPs (0, 10.84, 21.68 μg/mL). (2) Fasting food model (FFM) group: the initial food input was sterile water and the cells were incubated in complete 2 × DMEM containing the final small intestinal digesta and PS-NPs (0, 10.84, 21.68 μg/mL). (3) BSA group: the initial food input was BSA and the cells were incubated in complete 2 × DMEM containing the final small intestinal digesta and PS-NPs (0, 10.84, 21.68 μg/mL). (4) GT group: the initial food input was GT and the cells were incubated in complete 2 × DMEM containing the final small intestinal digesta and PS-NPs (0, 10.84, 21.68 μg/mL). (5) Starch group: the initial food input was starch and the cells were incubated in complete 2 × DMEM containing the final small intestinal digesta and PS-NPs (0, 10.84, 21.68 μg/mL). (6) Standard food model (SFM) group: the initial food input was a mixture of starch, BSA and GT, and the cells were incubated in complete 2 × DMEM containing the final small intestinal digesta and PS-NPs (0, 10.84, 21.68 μg/mL).

### 2.7. Cell viability

The Caco-2 cell viability was determined by the Cell Counting Kit-8 (CCK-8) method (Dojindo, Kyushu, Japan). The control and exposed cells were cultured at $1 \times 10^4$ cells/well in 96-well plates (Corning, NY, USA). Cell viability was determined every 24 h using six wells for each exposure concentration. We changed the cell exposure solution every day. After exposure, each well was washed with PBS. Then, 100 μL CCK-8 solution was added to each well of 96-well plates and incubated for 1 h. Readings were taken using a microplate reader (Biotek, Vermont, USA) under 450 nm and 630 nm settings. Cell survival was calculated based on absorbance.

### 2.8. Cellular uptake assay

We used GF PS-NPs in the cellular uptake assay. The fluorescence leakage from GF PS-NPs in digestive juices was negligible, which has been investigated in our previous study (Liang et al., 2021). The GF PS-NPs experienced *in vitro* gastrointestinal digestion with different food matrices, and the final digestas were applied to Caco-2 cells. We set 72 h exposure duration and changed the cell exposure solution daily based on our preliminary study. After exposure for 72 h, treatments were removed and the cells were washed three times with PBS, then incubated in complete DMEM and used to assess the cellular uptake of PS-NP. The images were acquired with a fluorescent inverted-microscope (Leica, Wetzlar, Germany) and the PS-NPs were visualized thanks to the green fluorophore they incorporate. The obtained images were analyzed and processed with ImageJ 1.52v (National Institute of Health, Bethesda, MD, USA).

### 2.9. Co-exposure to Cd and PS-NPs with mixture of food matrices

A growing body of literature has shifted focus from the co-exposure risks of MNPs to that of other food contaminants, e.g., monocrotophos (Mishra et al., 2022), lead (Wang et al., 2021b) and chromium (Jeong et al., 2022). We chose Cd, a common heavy metal contaminant in the food chain, to analyze its combined effect with NPs and food matrices. The Cd dosage in food input was set to 100 nM, which approximates the tolerable weekly intake and the mean intake in humans, and which reflects dietary exposure under real-world scenarios (Ba et al., 2017). To simulate human exposure conditions, Cd and a mixture of starch, BSA and GT, along with PS-NPs, acted as a food input. After *in vitro* gastrointestinal digestion, the final small intestinal digestas were diluted with complete 2 × DMEM and used to treat Caco-2 cells for 72 h. Thus, like the calculation process for doses of NP exposure in Caco-2 cells, the actual cell exposure concentration of Cd was 2.7 nM. To compare the cellular toxicities between co-exposure with and without *in vitro* gastrointestinal digestion, we also performed co-exposure to Cd and PS-NPs (Cd, 2.7 nM; PS-NPs, 10.84 μg/mL) without *in vitro* gastrointestinal digestion. All cell culture medium was changed every day.

### 2.10. RNA analysis

Total RNA of the Caco-2 cells was extracted using TRIZOL (Accurate Biology, Shanghai, China). For quantitative polymerase chain reaction (qPCR) analysis, the cDNA was synthesized using an Evo M-MLV One Step RT-PCR Kit (Accurate Biology) following the manufacturer's recommended protocols. The primers for RNA were designed and synthesized by Tsingke Biological Technology (Beijing, China). The qPCR was performed with a SYBR® Green Premix Pro Taq HS qPCR Kit (Accurate Biology) on the ABI QuantStudio™ 6 flex (Applied Biosystems, Carlsbad, USA). The qPCR conditions were as follows: 30 s at 95 °C, followed by 40 cycles of 5 s at 95 °C and 34 s at 60 °C. Primer specificity was confirmed by melting curve analysis, which showed a single product with the appropriate Tm for each primer set. All samples were analyzed in quadruplicate. The relative expression of all genes was normalized by *GAPDH*. Gene relative expression was determined with the $2^{-\Delta\Delta CT}$, compared with the control. The selected genes and their primer sequences are shown in Supplementary Information Table S1.

### 2.11. Statistical analysis

Data were expressed as mean ± standard deviation (SD). One-way analysis of variance followed by a Tukey multiple comparison test was used to compare the difference using SPSS 25.0 (SPSS Inc., Chicago, IL, USA). A probability ($P$) value $< 0.05$ was considered statistically significant. Statistical graphs were generated in Graph Pad Prism 8.0 (GraphPad Software, Inc., San Diego, CA, USA).

## 3. Results

### 3.1. Food matrices influenced PS-NP characterization

Colloidal characterization, including size, PDI and zeta-potential, were measured by DLS. The characterization of pristine PS-NPs and digested PS-NPs, as well as the final intestinal digestas diluted with 2 × DMEM, are summarized in Table 1. Without *in vitro* gastrointestinal digestion, both 10.84 and 21.68 μg/mL pristine NF PS-NPs showed similar particle size in distilled water. The particle size distribution was monodisperse for both 10.84 and 21.68 μg/mL pristine NF PS-NPs; they showed good stabilities in distilled water. The physical characteristics of pristine GF PS-NPs are similar to those of pristine NF PS-NPs. Conversely, the mean particle sizes of pristine NF PS-NPs diluted in complete 2 × DMEM (sizes = 208.3 nm for 10.84 μg/mL and 455.3 nm for 21.68 μg/mL) were different from those diluted in distilled water (size = 68.5 nm for 10.84 μg/mL and 61.0 nm for 21.68 μg/mL). Moreover, we observed elevated PDI and zeta potential after diluting pristine NF PS-NPs with complete 2 × DMEM. Food matrices' influence on the NF PS-NP characterization under *in vitro* gastrointestinal digestion was also examined. At 400 μg/mL, the PS-NPs' particle sizes in the FFM, starch, BSA, GT and SFM groups ranged from 61.7 to 67.7 nm. At 800 μg/mL, the PS-NPs' sizes in the FFM, starch, BSA and SFM groups ranged from 252.6 to 413.7 nm, whereas they were 53.7 nm in the GT group. We observed elevated PDI in the FFM, starch, BSA, GT and SFM groups at both 400 and 800 μg/mL. Meanwhile, we observed elevated zeta potential in the FFM, starch and BSA groups at 400 and 800 μg/mL. However, zeta potential was diminished in the SFM and GT groups. As shown by SEM, PS-NPs integrated with digesta and agglomerated after *in vitro* gastrointestinal digestion at both 400 and 800 μg/mL (Fig. 2). Taken together, these results demonstrated that PS-NP concentrations and various food matrices had influenced PS-NP characteristics during *in vitro* gastrointestinal digestion.

### 3.2. Food matrices altered the protein corona's protein composition on the PS-NP surface

To investigate the absorbed proteins on the NF PS-NP surface under *in vitro* gastrointestinal digestion, we analyzed protein content and the protein corona's protein composition. We found that protein corona formed on the PS-NP surface under *in vitro* gastrointestinal digestion,

**Table 1**
Physical characteristics of PS-NPs in different *in vitro* gastrointestinal digestions.

| Group | Dynamic Light Scattering | | |
|---|---|---|---|
| | Size (nm) | PDI | Zeta potential (mV) |
| Distilled water (NF, 10.84 μg/mL PS-NPs) | 55.7 ± 0.5 | 0.098 ± 0.054 | −22.4 ± 1.0 |
| Distilled water (NF, 21.68 μg/mL PS-NPs) | 61.0 ± 2.8 | 0.138 ± 0.056 | −23.4 ± 1.0 |
| Distilled water (GF, 10.84 μg/mL PS-NPs) | 53.4 ± 5.2 | 0.085 ± 0.006 | −25.8 ± 0.3 |
| Distilled water (GF, 21.68 μg/mL PS-NPs) | 54.7 ± 4.2 | 0.052 ± 0.012 | −25.6 ± 1.4 |
| DMEM (NF, 10.84 μg/mL PS-NPs) | 208.3 ± 26.2 | 0.620 ± 0.075 | −9.5 ± 0.5 |
| DMEM (NF, 21.68 μg/mL PS-NPs) | 455.3 ± 54.9 | 0.762 ± 0.015 | −9.8 ± 0.4 |
| DMEM (GF, 10.84 μg/mL PS-NPs) | 250.0 ± 25.4 | 0.775 ± 0.073 | −8.7 ± 0.5 |
| DMEM (GF, 21.68 μg/mL PS-NPs) | 493.8 ± 54.1 | 0.794 ± 0.084 | −7.4 ± 0.9 |
| a: Digesta (FFM + 400 μg/mL PS-NPs) | 67.7 ± 2.7 | 0.710 ± 0.098 | −15.5 ± 1.1 |
| b: Digesta (FFM + 800 μg/mL PS-NPs) | 413.7 ± 29.6 | 0.728 ± 0.128 | −17.3 ± 1.0 |
| a 1:3 DMEM (10.84 μg/mL PS-NPs) | 72.1 ± 16.7 | 0.724 ± 0.040 | −13.4 ± 0.9 |
| b 1:3 DMEM (21.68 μg/mL PS-NPs) | 381.8 ± 37.3 | 0.655 ± 0.089 | −15.5 ± 1.0 |
| c: Digesta (Starch + 400 μg/mL PS-NPs) | 61.7 ± 10.2 | 0.592 ± 0.030 | −16.2 ± 0.7 |
| d: Digesta (Starch + 800 μg/mL PS-NPs) | 342.5 ± 10.9 | 0.700 ± 0.083 | −15.2 ± 0.9 |
| c 1:3 DMEM (10.84 μg/mL PS-NPs) | 54.2 ± 5.2 | 0.679 ± 0.035 | −10.3 ± 0.4 |
| d 1:3 DMEM (21.68 μg/mL PS-NPs) | 395.9 ± 84.7 | 0.775 ± 0.073 | −8.2 ± 0.6 |
| e: Digesta (BSA + 400 μg/mL PS-NPs) | 67.7 ± 2.6 | 0.726 ± 0.057 | −19.4 ± 1.1 |
| f: Digesta (BSA + 800 μg/mL PS-NPs) | 261.0 ± 41.0 | 0.678 ± 0.069 | −14.3 ± 1.1 |
| e 1:3 DMEM (10.84 μg/mL PS-NPs) | 66.0 ± 8.6 | 0.794 ± 0.084 | −7.4 ± 0.9 |
| f 1:3 DMEM (21.68 μg/mL PS-NPs) | 497.5 ± 82.9 | 0.726 ± 0.047 | −8.7 ± 0.5 |
| g: Digesta (GT + 400 μg/mL PS-NPs) | 64.2 ± 9.5 | 0.751 ± 0.125 | −31.9 ± 0.4 |
| h: Digesta (GT + 800 μg/mL PS-NPs) | 53.7 ± 11.4 | 0.638 ± 0.126 | −30.9 ± 1.2 |
| g 1:3 DMEM (10.84 μg/mL PS-NPs) | 64.8 ± 14.5 | 0.557 ± 0.246 | −14.8 ± 1.5 |
| h 1:3 DMEM (21.68 μg/mL PS-NPs) | 54.2 ± 12.3 | 0.529 ± 0.009 | −18.0 ± 0.8 |
| i: Digesta (SFM + 400 μg/mL PS-NPs) | 65.7 ± 7.1 | 0.747 ± 0.050 | −25.0 ± 1.1 |
| j: Digesta (SFM + 800 μg/mL PS-NPs) | 252.6 ± 24.9 | 0.514 ± 0.069 | −29.7 ± 0.5 |
| i 1:3 DMEM (10.84 μg/mL PS-NPs) | 77.0 ± 2.1 | 0.537 ± 0.034 | −11.3 ± 1.3 |
| j 1:3 DMEM (21.68 μg/mL PS-NPs) | 111.1 ± 10.1 | 0.507 ± 0.026 | −16.2 ± 1.0 |

NF PS-NPs were used in *in vitro* gastrointestinal digestion and further characterization. BSA, bovine serum albumin; DMEM, Dulbecco's modified Eagle medium; FFM, fasting food model; GF, green fluorescence; PDI, polymer dispersion index; PS-NPs, polystyrene nanoplastics; SFM, standard food model; NF, non-fluorescence. $n = 3$ per group.

both with and without co-exposure with food matrices (Fig. 3A–E). Compared with the 10.84 μg/mL group, higher protein content was observed in the 21.68 μg/mL PS-NP, both with and without co-exposure to food matrices (Fig. 3B–E). Moreover, protein corona absorbed on PS-NPs comprised distinct proteins, both with and without co-exposure to food matrices, manifesting as diverse molecular weight distributions in the proteins of these groups (Fig. 3C–E). GT significantly changed the protein corona's protein composition on the PS-NP surface (Fig. 3E). However, we observed no significant difference in the protein corona's protein composition on PS-NP between 10.84 and 21.68 μg/mL groups, either with or without co-exposure to food matrices. These results implied that food matrices, especially GT, changed the protein corona's protein composition on the PS-NP surface.

### 3.3. Food matrices aggravated PS-NP-inhibited cell viability

We compared the altered Caco-2 cell viability caused by PS-NP exposure under various *in vitro* gastrointestinal digestion conditions (Fig. 4A–D). The results showed that cell viability had significantly decreased in both the FFM and the SFM groups, compared to the pristine group after 10.84 μg/mL PS-NPs exposure for 24 h (Fig. 4B). Additionally, after 21.68 μg/mL PS-NP exposure for 24 h, cell viability significantly decreased in the FFM, GT and SFM groups when compared to the pristine group (Fig. 4B). After exposure for 48 h, cell viability had significantly decreased, in both 10.84 and 21.68 μg/mL PS-NPs in the starch and GT groups, in comparison to the pristine group (Fig. 4C). After exposure for 72 h, cell viability had significantly decreased in both 10.84 and 21.68 μg/mL PS-NPs in the BSA, GT and SFM groups, while it had increased in the FFM group, compared to the pristine groups (Fig. 4D). Since co-exposing PS-NPs to either BSA, GT or SFM decreased cell viability in a dose-dependent manner at 72 h (Fig. 4D), we selected this time point in the following investigations. These results suggested that BSA and GT, as well as a mixture of starch, BSA and GT, exacerbated PS-NP-inhibited cell viability under *in vitro* gastrointestinal digestion. Among various food matrices, GT showed greater ability to aggravate PS-NP-inhibited cell viability.

### 3.4. Food matrices increased the Caco-2 cell uptake of PS-NPs

In order to observe whether PS-NPs could be internalized, we used GF PS-NPs to exam their uptake in the Caco-2 cells. After exposure to PS-NPs for 72 h, we detected green fluorescence inside the cells with various *in vitro* gastrointestinal digestion conditions (Fig. 5). We found Caco-2 cell uptake of PS-NPs in all PS-NP exposure groups (Fig. 5B–H). Fluorescence intensities significantly increased in 21.68 μg/mL PS-NPs, compared with corresponding 10.84 μg/mL PS-NPs in both the pristine group and the FFM and SFM groups (Fig. 5B, C, G, H). No significant changes in fluorescence intensities were found between 10.84 and 21.68 μg/mL PS-NP in the starch, BSA or GT groups (Fig. 5D, E, F, H). Fluorescence was significantly lower in the pristine PS-NP group than in the groups with *in vitro* gastrointestinal digestion, at both 10.84 and 21.68 μg/mL PS-NP concentrations (Fig. 5B–H). Furthermore, under *in vitro* gastrointestinal digestion, the cellular uptake of PS-NPs significantly increased at a 10.84 μg/mL PS-NP concentration in the groups with food matrices, compared with the group without food matrices (Fig. 5C–H). However, at a 21.68 μg/mL PS-NP concentration, the cellular uptake of PS-NPs increased only in the starch and BSA groups (Fig. 5D, E, H). These results revealed that gastrointestinal digestion had increased the cell uptake of PS-NPs, and food matrices had increased the Caco-2 cell uptake of PS-NP under *in vitro* gastrointestinal digestion.

### 3.5. Food matrices influenced PS-NP-altered expressions of genes correlated with inflammation, stress, and apoptosis

In order to assess whether various food matrices could change PS-NP-induced cellular toxicity under *in vitro* gastrointestinal digestion, we further analyzed the discrepancies in the PS-NP-altered mRNA levels of genes correlated with inflammation, stress and apoptosis in the Caco-2 cells (Fig. 6A-M). For inflammation-correlated genes, both *IL-6* or *IL-8* mRNA levels significantly decreased in the other groups, compared to the corresponding pristine group (Fig. 6C and D). The *IL-10* mRNA levels significantly decreased after exposure to 21.68 μg/mL PS-NPs in the FFM, starch, GT and SFM groups, compared to the pristine group. The *IL-1β* mRNA levels significantly increased after exposure to 21.68 μg/mL

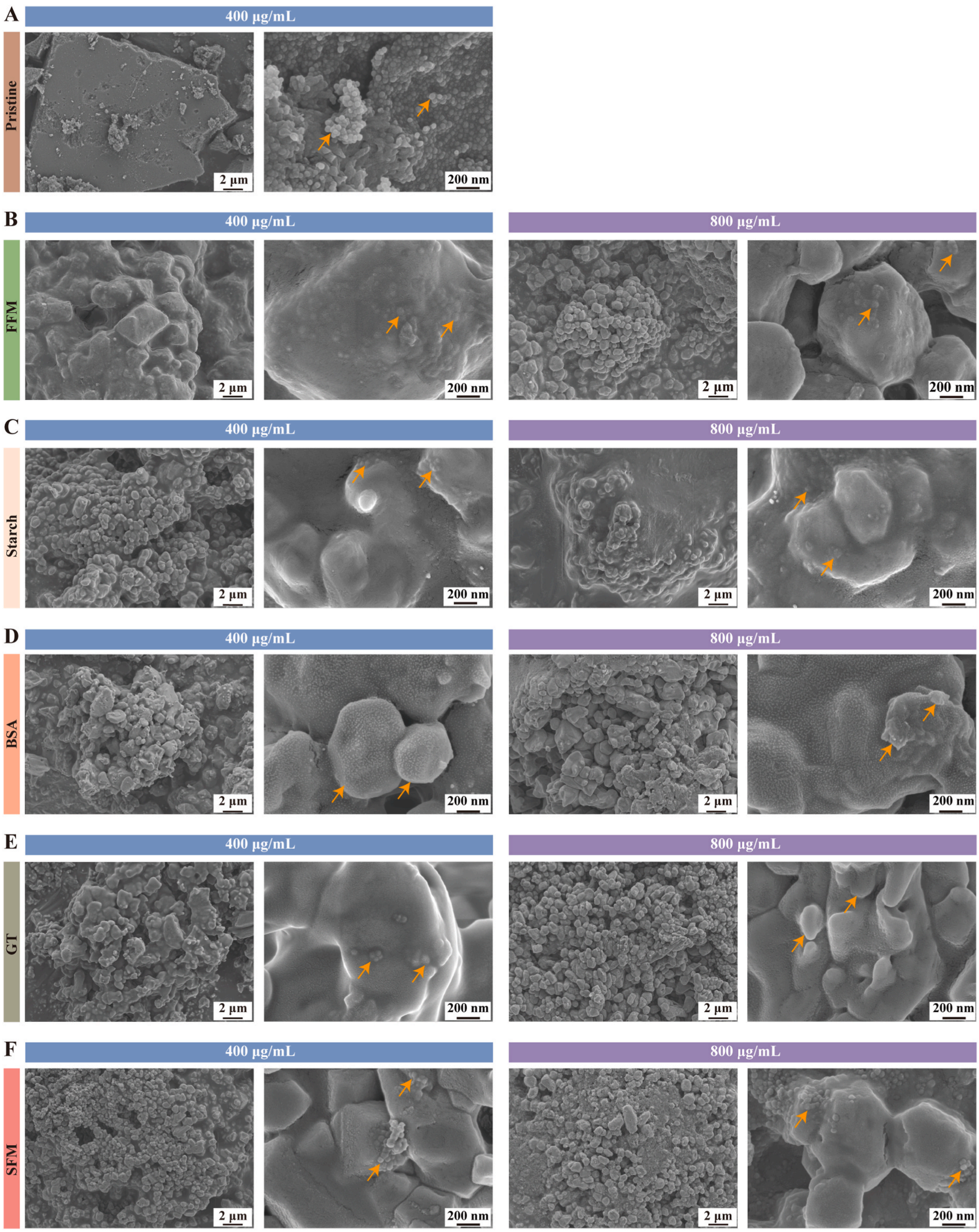


**Fig. 2.** Morphological characterization of PS-NPs under different *in vitro* gastrointestinal digestion conditions. Representative SEM images of 50 nm PS-NPs in groups of (A) pristine, (B) FFM, (C) Starch, (D) BSA, (E) GT and (F) SFM. BSA, bovine serum albumin; FFM, fasting food model; GT, glyceryl trioleate; NF, non-fluorescence; PS-NPs, polystyrene nanoplastics; SFM, standard food model. The yellow arrows indicate PS-NPs. Scale bar: 2 μm or 200 nm. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

PS-NPs in the FFM, GT and SFM, groups while they decreased in the BSA group, compared to the pristine group. For stress-correlated genes, PS-NP-decreased *HO1* and *SOD* mRNA levels were found in the starch and BSA groups, whereas increased *HO1* and *SOD* mRNA levels were found in the GT and SFM groups, compared to the pristine group (Fig. 6F and G). In contrast to the PS-NP-increased *HSP70* mRNA levels in the FFM and GT groups, the *HSP70* mRNA levels decreased in the BSA and SFM groups, when compared to the pristine group (Fig. 6I). PS-NPs-increased *GSTP1* mRNA levels were observed in the GT groups, compared to the pristine group. For apoptosis-correlated genes, PS-NPs increased the *BAX* and *CASP3* mRNA levels in the FFM group, but decreased them in the starch and BSA groups, relative to the pristine group (Fig. 6J, L). We also observed increased *CASP8* mRNA levels in the GT group, but decreased *CASP8* mRNA levels in the SFM group, when

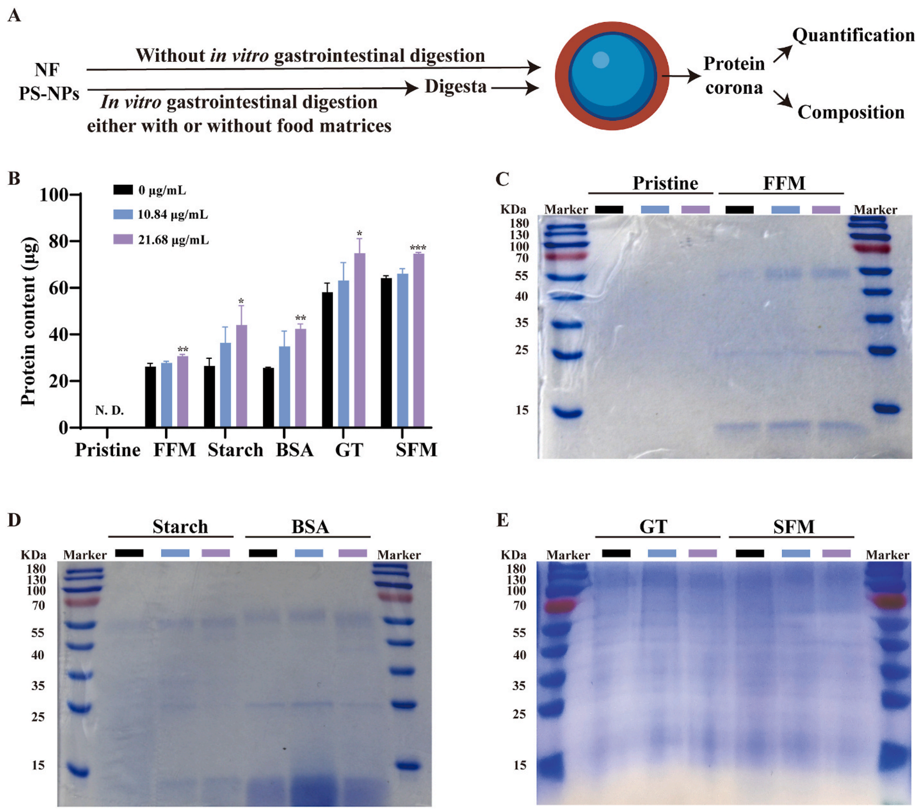


**Fig. 3.** Food matrices altered the absorbed protein patterns in the protein corona on the PS-NPs under different *in vitro* gastrointestinal digestion conditions. (A) Snapshot of the experimental design. (B) Protein quantification of the protein corona in each group. SDS-PAGE analysis of adsorbed proteins on the PS-NP surface in groups (C) pristine and FFM, (D) starch and BSA, and (E) GT and SFM. *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$ vs. without PS-NP exposure in each group. BSA, bovine serum albumin; FFM, fasting food model; GT, glyceryl trioleate; NF, non-fluorescence; PS-NPs, polystyrene nanoplastics; SFM, standard food model. N. D. refer to no detection.

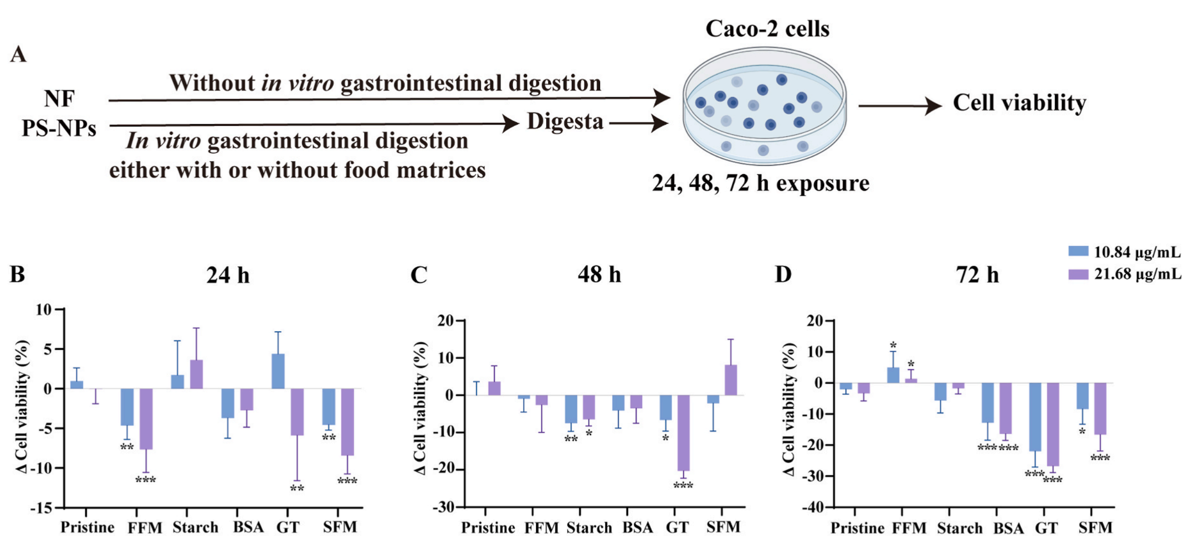


**Fig. 4.** Food matrices aggravated PS-NP-inhibited cell viability under *in vitro* gastrointestinal digestion. (A) Snapshot of the experimental design; (B) Δ Cell viability after 24 h exposure; (C) Δ Cell viability after 48 h exposure; (D) Δ Cell viability after 72 h exposure. Δ cell viability = (cell viability of NP group – cell viability of without NP group) in each *in vitro* gastrointestinal digestion condition. Values are expressed as the mean ± SD of three independent experiments. *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$ vs. the same concentration in the pristine PS-NP group. BSA, bovine serum albumin; FFM, fasting food model; GT, glyceryl trioleate; NF, non-fluorescence; PS-NPs: polystyrene nanoplastics; SFM, standard food model.

compared to the pristine group (Fig. 6M). Compared to the pristine group, *BCL2* mRNA levels decreased in the starch and GT groups. These findings showed that various food matrices in gastrointestinal digestion had influenced the PS-NP-altered gene expressions in the Caco-2 cells. Furthermore, both stress- and apoptosis-correlated genes changed in a dose-dependent manner (Fig. 6F-M), suggesting that co-exposure to food matrices with PS-NPs primarily affected stress and apoptosis in the Caco-2 cells. Specifically, BSA and GT showed greater ability to aggravate the PS-NP-altered mRNA expressions of stress- and apoptosis-correlated genes.

### *3.6. PS-NP and Cd co-exposure with a mixture of food matrices exhibited greater toxicity*

NPs generally coexist with other contaminants in food, and potentially induce combined effects (Mishra et al., 2022; Qiu et al., 2023). We

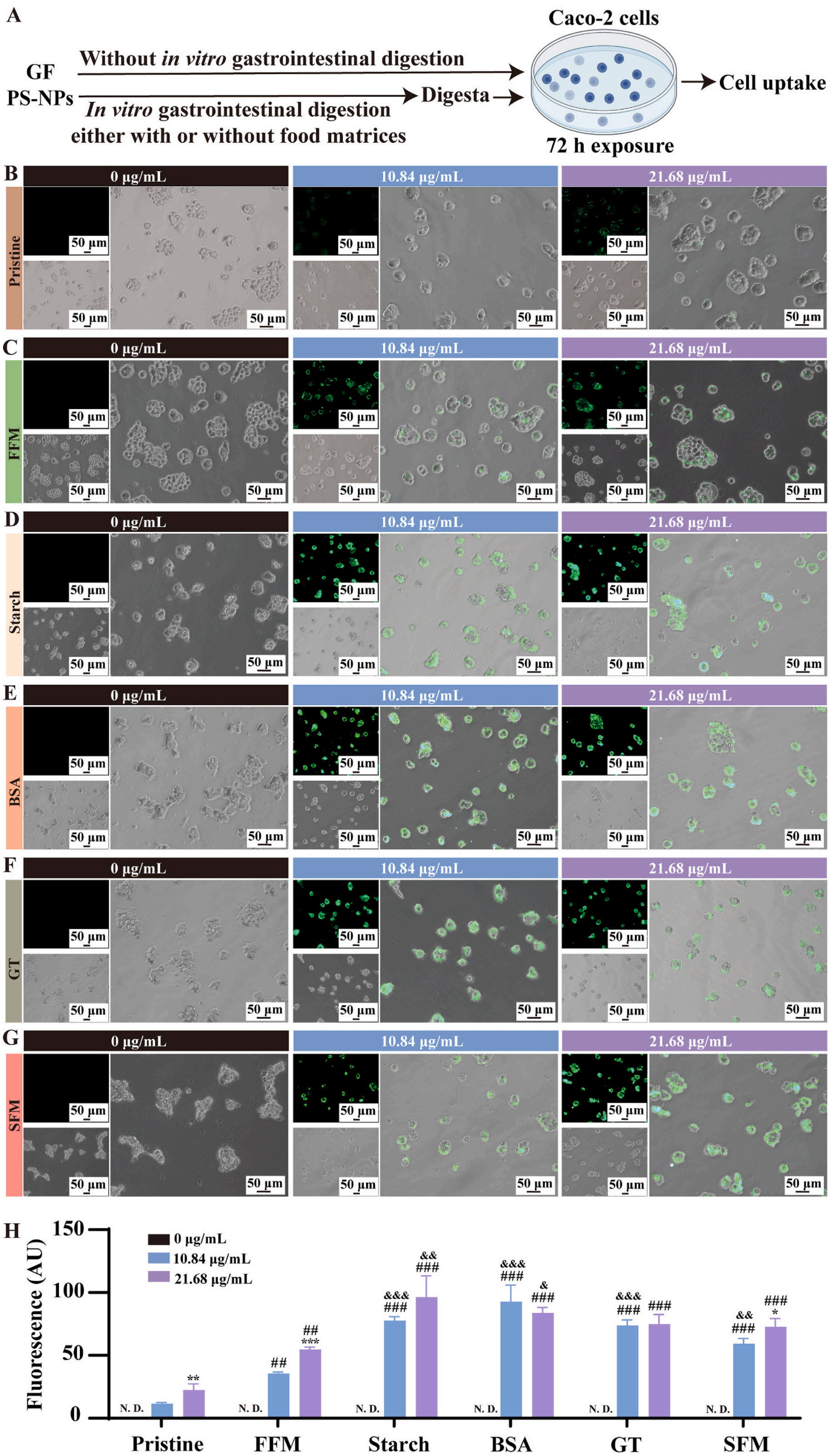


**Fig. 5.** GF PS-NP uptake in the Caco-2 cells under various *in vitro* gastrointestinal digestion conditions. (A) Snapshot of experiment design. GF PS-NP uptake in the (B) pristine GF PS-NP group, (C) FFM group, (D) starch group, (E) BSA group, (F) GT group, and (G) SFM group. (H) Fluorescence quantification in each group. The green fluorescence indicates GF PS-NPs. $^{*}P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$ vs. 10.84 μg/mL PS-NP concentration in each group. $\#P < 0.05$, $\#\#P < 0.01$, and $\#\#\#P < 0.001$ vs. the same concentration in the pristine PS-NP group. & $P < 0.05$, && $P < 0.01$, and &&& $P < 0.001$ vs. the same concentration in the FFM group. Scale bar: 50 μm. BSA, bovine serum albumin; FFM, fasting food model; GF, green fluorescence; GT, glyceryl trioleate; PS-NPs, polystyrene nanoplastics; SFM, standard food model. N. D. refer to no detection. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

investigated whether food matrices had ameliorated or exacerbated the combined effects of Cd and NP exposure. Herein, we evaluated the change in the expression of genes related to inflammation, stress and apoptosis after co-exposure to PS-NPs, Cd and food matrices under *in vitro* gastrointestinal digestion. We selected 10.84 μg/mL PS-NP and the mixture of starch, BSA and GT as food inputs to simulate the actual human exposure conditions for MNPs and foods (DeLoid et al., 2021; Laloux et al., 2020). Without *in vitro* gastrointestinal digestion, increased *HO1* and decreased *BCL2* mRNA levels were found after single Cd exposure, compared with the control group (Fig. 7F, K). We observed no change in the expressions of genes correlated with inflammation, stress or apoptosis when co-exposed to PS-NPs and Cd, compared with the

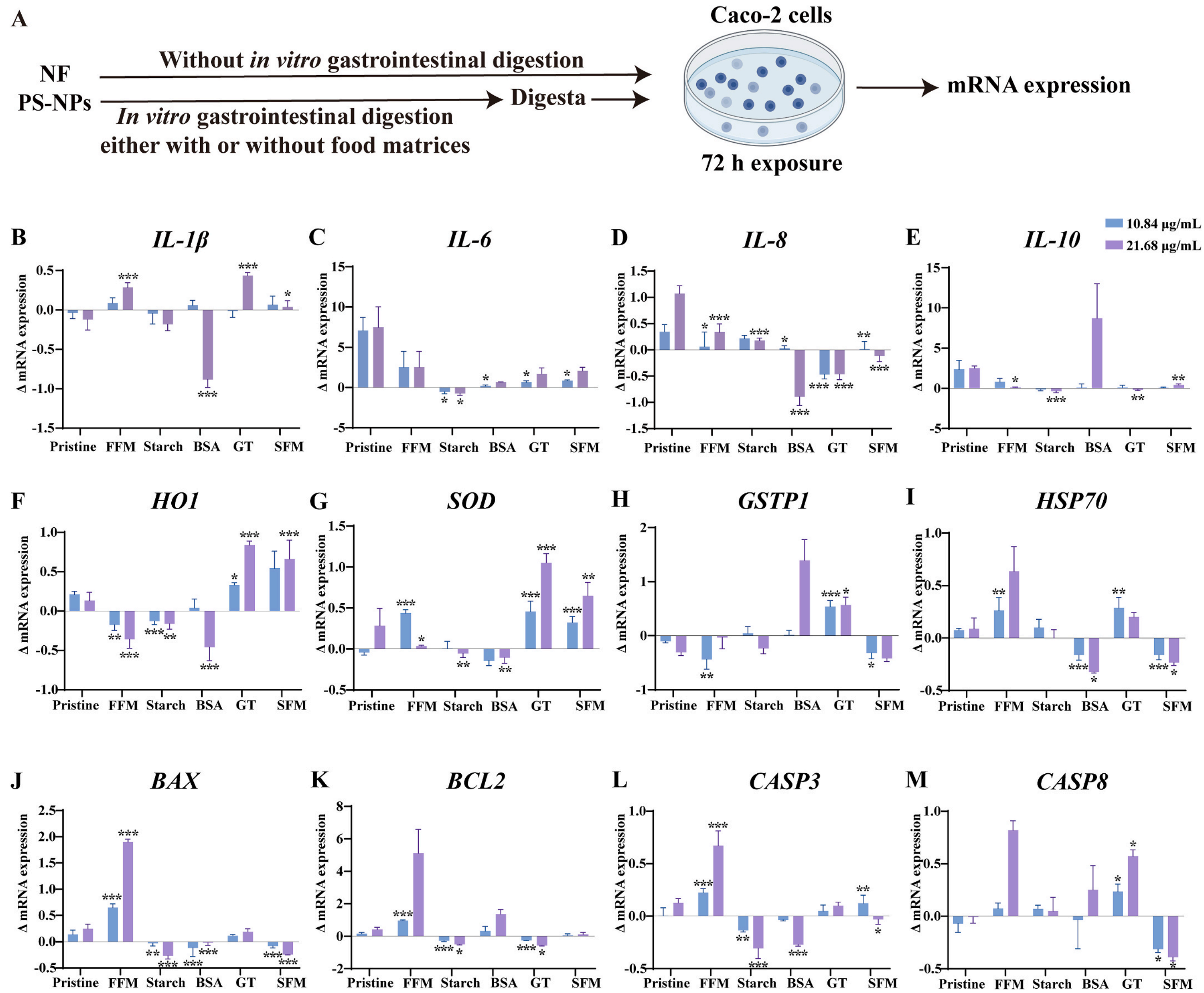


**Fig. 6.** Food matrices altered PS-NP-induced gene expression under various *in vitro* gastrointestinal digestion conditions. (A) Snapshot of the experimental design. Investigated inflammation-correlated gene list: (B) *IL-1β*, (C) *IL-6*, (D) *IL-8*, (E) *IL-10*; investigated stress-correlated gene list: (F) *HO1*, (G) *SOD*, (H) *GSTP1*, (I) *HSP70*; investigated apoptosis-correlated gene list: (J) *BAX*, (K) *BCL2*, (L) *CASP3*, and (M) *CASP8*. Δ mRNA expression = (relative mRNA expression of the NP group – relative mRNA expression of the without NP group) in each *in vitro* condition. Values are expressed as the mean ± SD of three independent experiments. $^*P < 0.05$, $^{**}P < 0.01$, and $^{***}P < 0.001$ vs. the same concentration in the pristine PS-NP group. BSA, bovine serum albumin; CASP, caspase; FFM, fasting food model; GSTP1, glutathione S-transferase P1; GT, glyceryl trioleate; HO1, heme oxygenase-1; HSP70, heat shock protein 70; IL, interleukin; NF, non-fluorescence; PS-NPs, polystyrene nanoplastics; SFM, standard food model; SOD, superoxidedismutase.

single Cd exposure group (Fig. 7B-M). With *in vitro* gastrointestinal digestion, we found that co-exposure to Cd and a mixture of food matrices significantly decreased *IL*-10 mRNA levels, whereas it increased *BAX* mRNA levels, compared with the control group. These results indicated that change in gene expression induced by co-exposure to food matrices and Cd under *in vitro* gastrointestinal digestion differed from that caused by single Cd exposure without *in vitro* gastrointestinal digestion. We also observed that co-exposure to PS-NPs and Cd with a mixture of food matrices increased *HO1* and *SOD* mRNA levels but decreased *GSTP1*, *HSP70*, *CASP8* and *BAX* mRNA levels, compared with the single Cd exposure group with *in vitro* gastrointestinal digestion (Fig. 7F, H, I, M). Collectively, these results suggested that gastrointestinal digestion with food matrices had exacerbated the negative effects of the cellular inflammation, stress and apoptosis due to co-exposure to NPs and Cd.

## 4. Discussion

Although MNPs have long been prevalent in the food chain, their interactions with varying food matrices and the resulting cellular toxicities remain unclear. We performed both physical characterization and toxicological evaluation to provide comprehensive insight into the effects of PS-NPs' interactions with various food matrices under *in vitro* gastrointestinal digestion. Our findings showed that PS-NPs absorbed proteins and formed agglomerations when they interacted with food matrices, and also when they did not. Combination with either BSA or GT, or a mixture of starch, BSA and GT, enhanced PS-NP-inhibited cell viability. In addition, food matrices aggravated the cellular toxicity of inflammation, stress and apoptosis induced by PS-NPs, as well as the combination of PS-NPs and Cd.

Passing through the gastrointestinal tract can alter nanoparticles' properties and influence their fate upon ingestion. Nanoparticles,

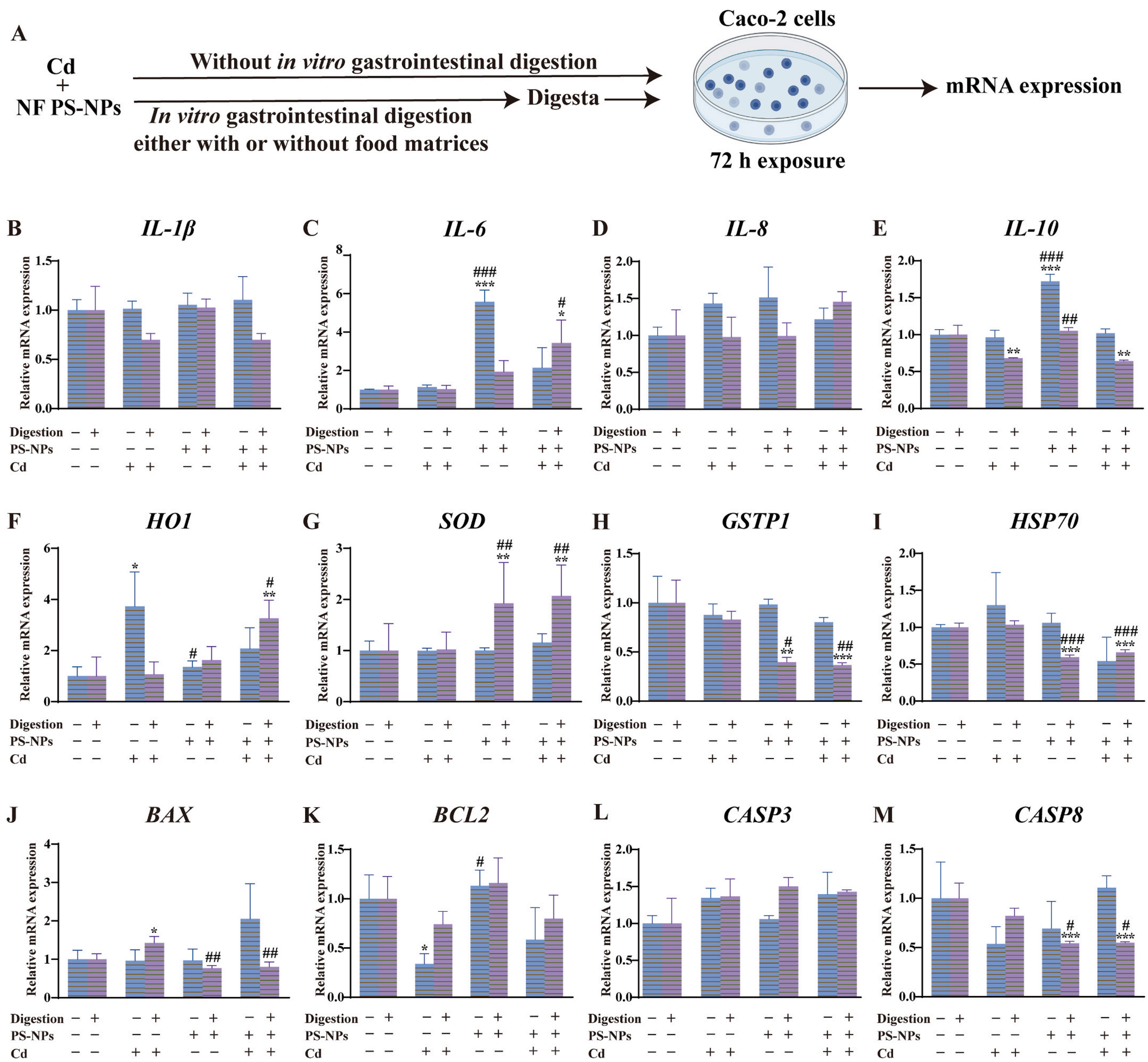


**Fig. 7.** Food matrices increased cellular toxicity induced by co-exposure to NPs and Cd under *in vitro* gastrointestinal digestion. (A) Snapshot of the experimental design. Investigated inflammation-correlated gene list: (B) *IL-1β*, (C) *IL-6*, (D) *IL-8*, (E) *IL-10*; investigated stress-correlated gene list: (F) *HO1*, (G) *SOD*, (H) *GSTP1*, (I) *HSP70*; investigated apoptosis-correlated gene list: (J) *BAX*, (K) *BCL2*, (L) *CASP3*, and (M) *CASP8*. Values are expressed as the mean ± SD of three independent experiments. *$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$ vs. the control group with neither PS-NP nor Cd exposure in the same *in vitro* condition. #$P < 0.05$, ##$P < 0.01$, and ###$P < 0.001$ vs. the single Cd exposure group in the same *in vitro* condition. CASP, caspase; Cd, cadmium; GSTP1, glutathione S-transferase P1; HO1, heme oxygenase-1; HSP70, heat shock protein 70; IL, interleukin; NF, non-fluorescence; PS-NPs, polystyrene nanoplastics; SOD, superoxidedismutase.

including silver (Laloux et al., 2020), ferric oxide (DeLoid et al., 2017) and titanium dioxide (Marucco et al., 2020), have been found to agglomerate in the gastric and small intestinal phase during *in vitro* gastrointestinal digestion. Furthermore, gastrointestinal passage can change PS-MPs' shape and size (Stock et al., 2020). PS-NPs become agglomerated in *in vitro* gastrointestinal tracts (Liu et al., 2020a). Nonetheless, whether and how varying food matrices influence the MNP characteristics and fate in the gastrointestinal tract remain unclear. Food matrices have been shown to change nanoparticle behaviors in the gastrointestinal tract (Zhou et al., 2021). Thus, we examined the effects of various food matrices upon NP features under *in vitro* gastrointestinal digestion. Our results showed that all selected food matrices, including starch, BSA, GT, and a mixture of them, aggravated PS-NP agglomeration. We also found that particle sizes were greater in the 21.68 μg/mL PS-NPs than that in the 10.84 μg/mL PS-NPs, excluding the GT group. Similarly, the PS-NP aggregation rate (faster aggregation and larger aggregates) increased along with mass concentration in the gastric system (Shao et al., 2022). We also found that the existence of GT increased the stability of corresponding final digestas, manifested as PS-NPs' decreased zeta potential in the GT and SFM groups. Consistently, corn oil has been found to decrease the zeta potential of simulated digestive fluids (Zhou et al., 2021). In sum, we found that PS-NPs aggregated during *in vitro* gastrointestinal digestion, both with and without a food matrix. Additionally, food matrices, i.e., lipids, can stabilize digestas.

Protein corona formed by nanoparticles absorb protein from the environment surrounding them and trigger nanoparticle aggregation (Ali et al., 2016; Latreille et al., 2022; Mishra et al., 2021). Protein corona formation occurs in several kinds of nanoparticles, including PS-NP (Coreas et al., 2020; Laloux et al., 2020; Liu et al., 2020a). It can contribute both benefits and toxicity to the nanoparticles (Mishra et al., 2021). For instance, protein corona accelerates drug delivery (Du et al., 2022; Zhang et al., 2019), and acts as a double-edged sword upon NP toxicity (Fadare et al., 2020). Prefabricated FBS protein corona mediates the relief of PS-NP-caused autophagic flux blockage, autophagosomes accumulation, and lysosomal damage in RAW264.7 cells (Tan et al., 2020b). In contrast, a growing body of literature has reported that protein corona increases nanoparticles' toxicity (Diaz-Diestra et al., 2022; Fadare et al., 2020; Rabel et al., 2021). Therefore, prior to PS-NPs' interaction with intestinal cells, protein corona formation on PS-NPs with various food matrices should be investigated. We observed that protein corona formed on the PS-NP surface under *in vitro* gastrointestinal digestion either with or without food matrices. Protein corona formation increased along with the increase in PS-NP concentrations. Furthermore, SDS-PAGE analysis demonstrated that protein composition in protein corona was different from co-exposure to PS-NPs and various food matrices. Several factors contribute to protein absorption, such as the hydrophilic/hydrophobic properties of nanoparticle surfaces and proteins' structural stability (Kopac, 2021). We inferred that food matrices altered protein corona by influencing protein compositions and the properties of the final digesta. Overall, food matrices impacted protein composition in protein corona on the PS-NP surface under *in vitro* gastrointestinal digestion.

Altered physical characteristics and absorbed protein corona have been reported to contribute to PS-NP-induced toxicity (Diaz-Diestra et al., 2022; Wang et al., 2021a). In the present study, we observed decreased cell viability in both BSA and GT and in SFM groups after 72 h PS-NP exposure. A recent study has demonstrated that cell viability decreases in the presence of 8.33 μg/mL MNPs, when compared to treatment with digestas from a high-fat food model alone (DeLoid et al., 2022). These results suggest that MNP toxicity is intensified under *in vitro* gastrointestinal digestion with lipids. In line with their results, we also found enhancement in PS-NP-inhibited cell viability under *in vitro* digestion with BSA, GT and SFM. Among them, GT aggravated PS-NP-inhibited cell viability to the highest degree. Absorption of both proteins and lipids on the NP surface can mediate NPs' interaction with cells (DeLoid et al., 2022). Therefore, we speculated that proteins and lipids are potential risk components of food regarding NP-induced toxicity. Nevertheless, how the lipid corona and protein corona affect NP toxicity remains underexplored. To address this concern, we investigated PS-NP cell intake in Caco-2 cells. We found that more PS-NPs (digested through the *in vitro* gastrointestinal tract, either with or without food matrices) reached the Caco-2 cells than the pristine PS-NPs. Similar results were found in a previous study, in which digested oil-free nanoparticles' absorption rate increased when compared with undigested nanoparticles (Chen et al., 2020). Furthermore, under *in vitro* gastrointestinal digestion, we found that the cell uptakes of PS-NPs in the groups with food matrices were greater than those in the group without food matrices. Collectively, our results implied that food matrix-altered protein corona composition promoted PS-NP cell uptake in the Caco-2 cells, resulting in decreased cell viability.

Most known mechanisms underlying PS-NP-induced intestinal toxicity are characterized by inflammation, stress and apoptosis (Domenech et al., 2021; He et al., 2022; Liang et al., 2021). Herein, we evaluated these potential mechanisms in Caco-2 cells treated by PS-NPs under *in vitro* gastrointestinal digestion with various food matrices. On the one hand, consistent with previous research (Liu et al., 2020a) in which digestive treatment increased PS-NP-induced proinflammatory effects, our results also showed that NPs increased cellular inflammation levels after *in vitro* gastrointestinal digestion. On the other hand, we observed greater stress and apoptosis levels under *in vitro* gastrointestinal digestion, both with and without food matrices, than in the pristine group. GT and BSA also caused greater stress levels than in the pristine group. This suggested that stress is one of the mechanisms underlying the aggravation of PS-NP toxicity under *in vitro* gastrointestinal digestion with GT and BSA. Overall, both inflammation and stress, as well as apoptosis caused by PS-NP exposure in the Caco-2 cells, varied with co-exposure to various food matrices.

The combined effects of MNPs and other food contaminants is an important aspect of MNP-aroused health concerns. MNPs intensify the toxicity of co-existing pollutants due to their capability to absorb them (Abduro Ogo et al., 2022; Liu et al., 2022). Because Cd contaminant is an important global concern in food safety (Xu et al., 2021), several studies have investigated the combined effects of Cd and MNPs (Chen et al., 2022; Liang et al., 2022b; Zuo et al., 2022). However, these studies have exclusively used non-mammalian models and high Cd concentrations from 5 μg/L to 2 mg/L (Cao et al., 2022; Luo et al., 2022; Wang et al., 2023). This is a far cry from real-world scenarios which are needed to accurately assess human risk due to co-exposure to MNPs and Cd. To fill these knowledge gaps, we investigated the combined effect of co-exposure to MNPs and Cd by using 2.7 nM Cd and 10.84 μg/mL PS-NPs in Caco-2 cells, both without *in vitro* gastrointestinal digestion and with *in vitro* gastrointestinal digestion via a mixture of food matrices. These selected food matrices and cellular concentrations of MNPs and Cd are similar to those that would be found in humans and in the environment. Without gastrointestinal digestion, we observed no significant changes in the selected gene levels in cells co-exposed to PS-NPs and Cd, compared with the single Cd exposure. However, with gastrointestinal digestion, co-exposure to PS-NPs and Cd with a mixture of food matrices increased *HO1* mRNA levels in the Caco-2 cells. HO1, a crucial metabolic enzyme, has emerged as a central effector of the mammalian stress response (Ryter, 2021). It exhibits low basal expression levels in most cells and tissues but is significantly upregulated by oxidative stress-inducing stimuli, such as heavy metals and lipopolysaccharide (Vijayan et al., 2010). Since both cadmium and PS-NPs are known to induce oxidative stress (Li et al., 2023; Liang et al., 2021), oxidative stress may play a key role in enhancing the toxicity of co-exposure to NPs and Cd, with HO1 potentially involved in the process. In addition, a previous study has demonstrated that *in vitro* gastrointestinal digestion significantly affects not only MNPs' properties, but also their adsorption capacity (Krasucka et al., 2022). After digestion, some functional groups may be introduced onto the surface of the PS-NPs, and thus alter their interactions with Cd (Krasucka et al., 2022; Liu et al., 2020b; Stock et al., 2020; Wang et al., 2021a). Thus, under gastrointestinal digestion with varying food matrices, PS-NPs may absorb more Cd, and thereafter induce more severe damage to the Caco-2 cells. Taken together, gastrointestinal digestion with varying food matrices could aggravate the combined effects of PS-NPs and Cd on cellular stress. This underscores the indispensable roles of the gastrointestinal tract and food matrices in human risk assessment for food MNP and Cd contaminant.

Some limitations in the present study should be noted. Firstly, we selected three representative food matrices, which is far less than the wealth of food in real-world scenarios. Secondly, we used 50 nm spherical PS particles as representative NPs. It is worth noting that both size and shape, as well as the material composition of NPs are major factors affecting MNP toxicity (Liang et al., 2021; Lozano et al., 2021; Stock et al., 2021). Therefore, more work is needed to clarify the interactions of MNPs of different sizes, shapes and materials with various food matrices. Thirdly, we preliminarily explored the mechanisms underlying the jeopardized Caco-2 cells by focusing on inflammation, stress and apoptosis. Thus, further studies warrant a comprehensive study of these mechanisms to dissect risk from NPs in the food chain.

## 5. Conclusion

This pioneering study compared the influence of food matrices on the characteristics and cellular toxicity of ingested NPs in simulated digestion. Food matrices affected PS-NPs' properties, aggregation, and absorbed protein corona. Cell uptake of PS-NPs was increased by food matrices, and BSA and GT worsened stress- and apoptosis-related gene expressions. Co-exposure to PS-NPs, Cd, and food matrices showed higher cellular toxicity. Our findings emphasize the significance of food components in MNPs and Cd risk assessment in the food chain.

## Funding

This work was supported by grants from the National Natural Science Foundation of China (Grant No. 82273656, 82073583, 82073519, 81872601), the Science and Technology Project of Guangdong Province, China (Grant No. 2022A0505050035), the Guangdong Basic Applied Basic Research Foundation (Grant No. 2022A1515010610, Guangdong-Dongguan Joint grant No. 2022A1515111098), the Chinese Postdoctoral Science Foundation (Grant No. 2022M721486), the Guangdong Key R&D Program Ministry of Science and Technology of Guangdong Province (Grant No. 2019B020210002), the National Training Program of Innovation and Entrepreneurship for Undergraduates (Grant No. S202212121034, S202112121113, S202112121115), the Guangdong Provincial Key Laboratory of Tropical Disease Research (Grant No. 2017B030314035) and the NMPA Key Laboratory for Safety Evaluation of Cosmetics.

## CRediT authorship contribution statement

**Zhiming Li:** Conceptualization, Methodology, Visualization, Data curation, Formal analysis, Writing – original draft. **Yuji Huang:** Data curation. **Yizhou Zhong:** Data curation. **Boxuan Liang:** Data curation. **Xingfen Yang:** Project administration. **Qing Wang:** Project administration. **Haixia Sui:** Funding acquisition, Resources. **Zhenlie Huang:** Conceptualization, Resources, Funding acquisition, Project administration, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

## Data availability

Data will be made available on request.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fct.2023.113984.

## References


Abduro Ogo, H., Tang, N., Li, X., Gao, X., Xing, W., 2022. Combined toxicity of microplastic and lead on submerged macrophytes. Chemosphere 295, 133956.

Ali, N., Mattsson, K., Rissler, J., Karlsson, H.M., Svensson, C.R., Gudmundsson, A., Lindh, C.H., Jönsson, B.A.G., Cedervall, T., Kåredal, M., 2016. Analysis of nanoparticle-protein coronas formed in vitro between nanosized welding particles and nasal lavage proteins. Nanotoxicology 10, 226–234.

Amato-Lourenco, L.F., Carvalho-Oliveira, R., Junior, G.R., Dos Santos Galvao, L., Ando, R.A., Mauad, T., 2021. Presence of airborne microplastics in human lung tissue. J. Hazard Mater. 416, 126124.

Ba, Q., Li, M., Chen, P., Huang, C., Duan, X., Lu, L., Li, J., Chu, R., Xie, D., Song, H., Wu, Y., Ying, H., Jia, X., Wang, H., 2017. Sex-dependent effects of cadmium exposure in early life on gut microbiota and fat accumulation in mice. Environ. Health Perspect. 125 (3), 437–446.

Belitz, H.D., Grosch, W., Schieberle, P., 2009. Food Chemistry, fourth ed. Springer, Berlin, Germany.

Bove, P., Malvindi, M.A., Kote, S.S., Bertorelli, R., Summa, M., Sabella, S., 2017. Dissolution test for risk assessment of nanoparticles: a pilot study. Nanoscale 9 (19), 6315–6326.

Cao, J., Liao, Y., Yang, W., Jiang, X., Li, M., 2022. Enhanced microalgal toxicity due to polystyrene nanoplastics and cadmium co-exposure: from the perspective of physiological and metabolomic profiles. J. Hazard Mater. 427, 127937.

Chen, L., Yokoyama, W., Liang, R., Tam, C., Miller, J., Zhong, F., 2020. Remodeling of β-carotene-encapsulated protein-stabilized nanoparticles during gastrointestinal digestion in vitro and in mice. J. Agric. Food Chem. 68 (52), 15468–15477.

Chen, X., Wang, J., Xie, Y., Ma, Y., Zhang, J., Wei, H., Abdou, A.I.E., 2022. Physiological response and oxidative stress of grass carp (Ctenopharyngodon idellus) under single and combined toxicity of polystyrene microplastics and cadmium. Ecotoxicol. Environ. Saf. 245, 114080.

Coreas, R., Cao, X., Deloid, G.M., Demokritou, P., Zhong, W., 2020. Lipid and protein corona of food-grade TiO2 nanoparticles in simulated gastrointestinal digestion. NanoImpact 20, 100272.

Cox, K.D., Covernton, G.A., Davies, H.L., Dower, J.F., Juanes, F., Dudas, S.E., 2019. Human consumption of microplastics. Environ. Sci. Technol. 53 (12), 7068–7074.

DeLoid, G.M., Cao, X., Bitounis, D., Singh, D., Llopis, P.M., Buckley, B., Demokritou, P., 2021. Toxicity, uptake, and nuclear translocation of ingested micro-nanoplastics in an in vitro model of the small intestinal epithelium. Food Chem. Toxicol. 158, 112609.

DeLoid, G.M., Cao, X., Coreas, R., Bitounis, D., Singh, D., Zhong, W., Demokritou, P., 2022. Incineration-generated polyethylene micro-nanoplastics increase triglyceride lipolysis and absorption in an in vitro small intestinal epithelium model. Environ. Sci. Technol. 56 (17), 12288–12297.

DeLoid, G.M., Wang, Y., Kapronezai, K., Lorente, L.R., Zhang, R., Pyrgiotakis, G., Konduru, N.V., Ericsson, M., White, J.C., De La Torre-Roche, R., Xiao, H., McClements, D.J., Demokritou, P., 2017. An integrated methodology for assessing the impact of food matrix and gastrointestinal effects on the biokinetics and cellular toxicity of ingested engineered nanomaterials. Part. Fibre Toxicol. 14 (1), 40.

Diaz-Diestra, D.M., Palacios-Hernandez, T., Liu, Y., Smith, D.E., Nguyen, A.K., Todorov, T., Gray, P.J., Zheng, J., Skoog, S.A., Goering, P.L., 2022. Impact of surface chemistry of ultrasmall superparamagnetic iron oxide nanoparticles on protein corona formation and endothelial cell uptake, toxicity, and barrier function. Toxicol. Sci. 188 (2), 261–275.

Domenech, J., de Britto, M., Velazquez, A., Pastor, S., Hernandez, A., Marcos, R., Cortes, C., 2021. Long-Term effects of polystyrene nanoplastics in human intestinal caco-2 cells. Biomolecules 11 (10), 1442.

Du, Y., Zhou, H., Su, G., Ma, M., Liu, Y., 2022. Protein corona-driven nanovaccines improve antigen intracellular release and immunotherapy efficacy. Journal of Control 345, 601–609. Release.

Fadare, O.O., Wan, B., Liu, K., Yang, Y., Zhao, L., Guo, L.H., 2020. Eco-corona vs protein corona: effects of humic substances on corona formation and nanoplastic particle toxicity in Daphnia magna. Environ. Sci. Technol. 54 (13), 8001–8009.

Gautam, R., Jo, J., Acharya, M., Maharjan, A., Lee, D., K, C.P., Kim, C., Kim, K., Kim, H., Heo, Y., 2022. Evaluation of potential toxicity of polyethylene microplastics on human derived cell lines. Sci. Total Environ. 838 (Pt 2), 156089.

Hardy, A., Benford, D., Halldorsson, T., Jeger, M.J., Knutsen, H.K., More, S., Naegeli, H., Noteborn, H., Ockleford, C., Ricci, A., Rychen, G., Schlatter, J.R., Silano, V., Solecki, R., Turck, D., Younes, M., Chaudhry, Q., Cubadda, F., Gott, D., Oomen, A., Weigel, S., Karamitrou, M., Schoonjans, R., Mortensen, A., 2018. Guidance on risk assessment of the application of nanoscience and nanotechnologies in the food and feed chain: Part 1, human and animal health. EFSA J. 16, e05327.

He, Y., Li, Z., Xu, T., Luo, D., Chi, Q., Zhang, Y., Li, S., 2022. Polystyrene nanoplastics deteriorate LPS-modulated duodenal permeability and inflammation in mice via ROS drived-NF-κB/NLRP3 pathway. Chemosphere 307 (Pt 1), 135662.

Hollebeeck, S., Borlon, F., Schneider, Y.J., Larondelle, Y., Rogez, H., 2013. Development of a standardised human in vitro digestion protocol based on macronutrient digestion using response surface methodology. Food Chem. 138 (2–3), 1936–1944.

Jeong, H., Lee, Y.H., Sayed, A.E.D.H., Jeong, C.B., Zhou, B., Lee, J.S., Byeon, E., 2022. Short- and long-term single and combined effects of microplastics and chromium on the freshwater water flea Daphnia magna. Aquat. Toxicol. 253, 106348.

Kopac, T., 2021. Protein corona, understanding the nanoparticle-protein interactions and future perspectives: a critical review. Int. J. Biol. Macromol. 169, 290–301.

Krasucka, P., Bogusz, A., Baranowska-Wojcik, E., Czech, B., Szwajgier, D., Rek, M., Ok, Y. S., Oleszczuk, P., 2022. Digestion of plastics using in vitro human gastrointestinal tract and their potential to adsorb emerging organic pollutants. Sci. Total Environ. 843, 157108.

Laloux, L., Kastrati, D., Cambier, S., Gutleb, A.C., Schneider, Y.J., 2020. The food matrix and the gastrointestinal fluids alter the features of silver nanoparticles. Small 16 (21), e1907687.

Latreille, P.L., Rabanel, J.M., Le Goas, M., Salimi, S., Arlt, J., Patten, S.A., Ramassamy, C., Hildgen, P., Martinez, V.A., Banquy, X., 2022. In situ characterization of the protein corona of nanoparticles in vitro and in vivo. Adv. Mater. 34 (38), e2203354.

Leslie, H.A., van Velzen, J.M., Brandsma, S.H., Vethaak, D., Garcia-Vallejo, J.J., Lamoree, M.H., 2022. Discovery and quantification of plastic particle pollution in human blood. Environ. Int. 163, 107199.

Li, Z., Feng, C., Pang, W., Tian, C., Zhao, Y., 2021. Nanoplastic-induced genotoxicity and intestinal damage in freshwater benthic clams (corbicula fluminea): comparison with microplastics. ACS Nano 15 (6), 9469–9481.

Li, Z., Liao, W., Yin, X., Liu, L., Zhao, Z., Lu, X., Xu, F., Lin, X., Chen, Y., Song, J., He, Z., Wei, Q., Wu, W., Wu, Y., Yang, X., 2023. Hyperoside attenuates Cd-induced kidney injury via inhibiting NLRP3 inflammasome activation and ROS/MAPK/NF-κB signaling pathway in vivo and in vitro. Food Chem. Toxicol. 172, 113601.

Liang, B., Huang, Y., Zhong, Y., Li, Z., Ye, R., Wang, B., Zhang, B., Meng, H., Lin, X., Du, J., Hu, M., Wu, Q., Sui, H., Yang, X., Huang, Z., 2022a. Brain single-nucleus transcriptomics highlights that polystyrene nanoplastics potentially induce

Parkinson's disease-like neurodegeneration by causing energy metabolism disorders in mice. J. Hazard Mater. 430, 128459.
Liang, B., Zhong, Y., Huang, Y., Lin, X., Liu, J., Lin, L., Hu, M., Jiang, J., Dai, M., Wang, B., Zhang, B., Meng, H., Lelaka, J.J.J., Sui, H., Yang, X., Huang, Z., 2021. Underestimated health risks: polystyrene micro- and nanoplastics jointly induce intestinal barrier dysfunction by ROS-mediated epithelial cell apoptosis. Part. Fibre Toxicol. 18 (1), 20.
Liang, X., Zhou, D., Wang, J., Li, Y., Liu, Y., Ning, Y., 2022b. Evaluation of the toxicity effects of microplastics and cadmium on earthworms. Sci. Total Environ. 836, 155747.
Lichtenstein, D., Ebmeyer, J., Knappe, P., Juling, S., Bohmert, L., Selve, S., Niemann, B., Braeuning, A., Thunemann, A.F., Lampen, A., 2015. Impact of food components during in vitro digestion of silver nanoparticles on cellular uptake and cytotoxicity in intestinal cells. Biol. Chem. 396 (11), 1255–1264.
Liu, S., Wu, X., Gu, W., Yu, J., Wu, B., 2020a. Influence of the digestive process on intestinal toxicity of polystyrene microplastics as determined by in vitro Caco-2 models. Chemosphere 256, 127204.
Liu, X., Gharasoo, M., Shi, Y., Sigmund, G., Hüffer, T., Duan, L., Wang, Y., Ji, R., Hofmann, T., Chen, W., 2020b. Key physicochemical properties dictating gastrointestinal bioaccessibility of microplastics-associated organic xenobiotics: insights from a deep learning approach. Environ. Sci. Technol. 54 (19), 12051–12062.
Liu, X., Ma, J., Guo, S., Shi, Q., Tang, J., 2022. The combined effects of nanoplastics and dibutyl phthalate on Streptomyces coelicolor M145. Sci. Total Environ. 826, 154151.
Lozano, Y.M., Lehnert, T., Linck, L.T., Lehmann, A., Rillig, M.C., 2021. Microplastic shape, polymer type, and concentration affect soil properties and plant biomass. Front. Plant Sci. 12, 616645.
Luo, B., Li, J., Wang, M., Zhang, X., Mi, Y., Xiang, J., Gong, S., Zhou, Y., Ma, T., 2022. Chronic toxicity effects of sediment-associated polystyrene nanoplastics alone and in combination with cadmium on a keystone benthic species Bellamya aeruginosa. J. Hazard Mater. 433, 128800.
Marucco, A., Prono, M., Beal, D., Alasonati, E., Fisicaro, P., Bergamaschi, E., Carriere, M., Fenoglio, I., 2020. Biotransformation of food-grade and nanometric TiO2 in the oral-gastro-intestinal tract: driving forces and effect on the toxicity toward intestinal epithelial cells. Nanomaterials 10 (11), 2132.
Mishra, C.S.K., Samal, S., Samal, R.R., Behera, B.P., Pallavini, P., Dash, P., Brahma, S., Moharana, T., Pradhan, S., Acharya, P., 2022. Polyvinylchloride and polypropylene as adsorbents of the pesticide monocrotophos enhance oxidative stress in Eudrillus eugeniae (Kinberg). Chemosphere 295, 133837.
Mishra, R.K., Ahmad, A., Vyawahare, A., Alam, P., Khan, T.H., Khan, R., 2021. Biological effects of formation of protein corona onto nanoparticles. Int. J. Biol. Macromol. 175, 1–18.
Pironti, C., Ricciardi, M., Motta, O., Miele, Y., Proto, A., Montano, L., 2021. Microplastics in the environment: intake through the food web, human exposure and toxicological effects. Toxics 9 (9), 224.
Qiaorun, Z., Honghong, S., Yao, L., Bing, J., Xiao, X., Julian McClements, D., Chongjiang, C., Biao, Y., 2022. Investigation of the interactions between food plant carbohydrates and titanium dioxide nanoparticles. Food Res. Int. 159, 111574.
Qiu, W., Ye, J., Su, Y., Zhang, X., Pang, X., Liao, J., Wang, R., Zhao, C., Zhang, H., Hu, L., Tang, Z., Su, R., 2023. Co-exposure to environmentally relevant concentrations of cadmium and polystyrene nanoplastics induced oxidative stress, ferroptosis and excessive mitophagy in mice kidney. Environ. Pollut., 121947
Rabel, M., Warncke, P., Thurmer, M., Gruttner, C., Bergemann, C., Kurland, H.D., Muller, F.A., Koeberle, A., Fischer, D., 2021. The differences of the impact of a lipid and protein corona on the colloidal stability, toxicity, and degradation behavior of iron oxide nanoparticles. Nanoscale 13 (20), 9415–9435.
Ragusa, A., Svelato, A., Santacroce, C., Catalano, P., Notarstefano, V., Carnevali, O., Papa, F., Rongioletti, M.C.A., Baiocco, F., Draghi, S., D'Amore, E., Rinaldo, D., Matta, M., Giorgini, E., 2021. Plasticenta: first evidence of microplastics in human placenta. Environ. Int. 146, 106274.
Ryter, S.W., 2021. Heme oxgenase-1, a cardinal modulator of regulated cell death and inflammation. Cells 10 (3), 515.
Senathirajah, K., Attwood, S., Bhagwat, G., Carbery, M., Wilson, S., Palanisami, T., 2021. Estimation of the mass of microplastics ingested - a pivotal first step towards human health risk assessment. J. Hazard Mater. 404 (Pt B), 124004.
Shao, Z., Su, J., Dong, J., Liang, M., Xiao, J., Liu, J., Zeng, Q., Li, Y., Huang, W., Chen, C., 2022. Aggregation kinetics of polystyrene nanoplastics in gastric environments: effects of plastic properties, solution conditions, and gastric constituents. Environ. Int. 170, 107628.
Stock, V., Fahrenson, C., Thuenemann, A., Donmez, M.H., Voss, L., Bohmert, L., Braeuning, A., Lampen, A., Sieg, H., 2020. Impact of artificial digestion on the sizes and shapes of microplastic particles. Food Chem. Toxicol. 135, 111010.
Stock, V., Laurisch, C., Franke, J., Donmez, M.H., Voss, L., Bohmert, L., Braeuning, A., Sieg, H., 2021. Uptake and cellular effects of PE, PP, PET and PVC microplastic particles. Toxicol. Vitro 70, 105021.
Tan, H., Yue, T., Xu, Y., Zhao, J., Xing, B., 2020a. Microplastics reduce lipid digestion in simulated human gastrointestinal system. Environ. Sci. Technol. 54 (19), 12285–12294.
Tan, Y., Zhu, X., Wu, D., Song, E., Song, Y., 2020b. Compromised autophagic effect of polystyrene nanoplastics mediated by protein corona was recovered after lysosomal degradation of corona. Environ. Sci. Technol. 54 (18), 11485–11493.
Tolardo, V., Magrì, D., Fumagalli, F., Cassano, D., Athanassiou, A., Fragouli, D., Gioria, S., 2022. In vitro high-throughput toxicological assessment of nanoplastics. Nanomaterials 12 (12), 1947.
Vijayan, V., Mueller, S., Baumgart-Vogt, E., Immenschuh, S., 2010. Heme oxygenase-1 as a therapeutic target in inflammatory disorders of the gastrointestinal tract. World J. Gastroenterol. 16 (25), 3112–3119.
Wang, L., Wang, Y., Xu, M., Ma, J., Zhang, S., Liu, S., Wang, K., Tian, H., Cui, J., 2021a. Enhanced hepatic cytotoxicity of chemically transformed polystyrene microplastics by simulated gastric fluid. J. Hazard Mater. 410, 124536.
Wang, Q., Wang, J., Chen, H., Zhang, Y., 2023. Toxicity effects of microplastics and nanoplastics with cadmium on the alga Microcystis aeruginosa. Environ. Sci. Pollut. Res. Int. 30 (7), 17360–17373.
Wang, S., Li, Q., Huang, S., Zhao, W., Zheng, Z., 2021b. Single and combined effects of microplastics and lead on the freshwater algae Microcystis aeruginosa. Ecotoxicol. Environ. Saf. 208, 111664.
Wu, B., Wu, X., Liu, S., Wang, Z., Chen, L., 2019. Size-dependent effects of polystyrene microplastics on cytotoxicity and efflux pump inhibition in human Caco-2 cells. Chemosphere 221, 333–341.
Wu, D., Feng, Y., Wang, R., Jiang, J., Guan, Q., Yang, X., Wei, H., Xia, Y., Luo, Y., 2022. Pigment microparticles and microplastics found in human thrombi based on Raman spectral evidence. J. Adv. Res. 49, 141–150.
Wu, S., Wu, M., Tian, D., Qiu, L., Li, T., 2020. Effects of polystyrene microbeads on cytotoxicity and transcriptomic profiles in human Caco-2 cells. Environ. Toxicol. 35 (4), 495–506.
Xu, F.F., Song, J., Li, Y.Q., Lai, Y.F., Lin, J., Pan, J.L., Chi, H.Q., Wang, Y., Li, Z.Y., Zhang, G.Q., Cai, Z.F., Liang, X.X., Ma, A.D., Tan, C.T., Wu, W.L., Yang, X.F., 2021. Bioaccessibility and bioavailability adjusted dietary exposure of cadmium for local residents from a high-level environmental cadmium region. J. Hazard Mater. 420, 126550.
Yan, Z., Liu, Y., Zhang, T., Zhang, F., Ren, H., Zhang, Y., 2021. Analysis of microplastics in human feces reveals a correlation between fecal microplastics and inflammatory bowel disease status. Environ. Sci. Technol. 56 (1), 414–421.
Zhang, N., Li, Y.B., He, H.R., Zhang, J.F., Ma, G.S., 2021. You are what you eat: microplastics in the feces of young men living in Beijing. Sci. Total Environ. 767, 144345.
Zhang, Z., Guan, J., Jiang, Z., Yang, Y., Liu, J., Hua, W., Mao, Y., Li, C., Lu, W., Qian, J., Zhan, C., 2019. Brain-targeted drug delivery by manipulating protein corona functions. Nat. Commun. 10 (1), 3561.
Zhou, P., Guo, M., Cui, X., 2021. Effect of food on orally-ingested titanium dioxide and zinc oxide nanoparticle behaviors in simulated digestive tract. Chemosphere 268, 128843.
Zuo, Z., Wang, Q., Zhang, C., Zou, J., 2022. Single and combined effects of microplastics and cadmium on juvenile grass carp (Ctenopharyngodon idellus). Comp. Biochem. Physiol. C Toxicol. Pharmacol. 261, 109424.